Noise

Noise is a random process. The value of noise cannot be predicted at any time if the past values are known. Noise is a statistical in nature, represented by power spectral density $S_{xx}(f)$. The spectrum shows how much power the signal carriers at each frequency. The PSD, $S_x(f)$, of a noise waveform $x(t)$ is defined as the average power carried by $x(t)$ in a one-hertz bandwidth f. $S_x(f)$ is expressed in V^2/Hz. It is also common to take the square root of $S_x(f)$, expressing the result in $V/\sqrt{Hz}$.

For example, the input noise voltage of an amplifier at 100 MHz is equal to $3nV/\sqrt{Hz}$, simply to mean that the average power in a 1-Hz bandwidth at 100 MHz is equal to $(3\times10^{-9})^2 V^2$.

An example of common type of noise PSD is the "White spectrum," also called white noise. Shown in Fig. 1, such a PSD displays the same value at all frequencies (Similar to white light).

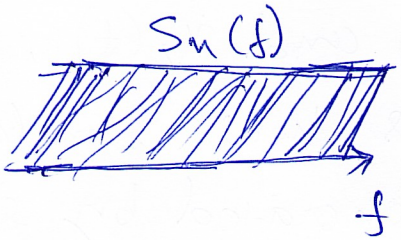


Fig. 1 White ~~no~~ spectrum

The white noise does not exist because the ~~total~~ area under the PSD, i.e the ~~total~~ power carried by the noise, is infinite. In practice, however, any noise spectrum that is flat in the

band of interest is usually called white.

Correlated and Uncorrelated Sources: In noise analysis, the average noise power is of interest, we add two noise waveforms and take the average of the resulting power:

$$P_{av} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{+T/2} [x_1(t) + x_2(t)]^2 \, dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{+T/2} x_1^2(t)\, dt + \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{+T/2} x_2^2(t)\, dt$$

$$+ \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{+T/2} 2\, x_1(t)\, x_2(t)\, dt$$

$$= P_{av_1} + P_{av_2} + \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{+T/2} 2\, x_1(t)\, x_2(t)\, dt \quad \text{---} (1)$$

where P_{av_1} and P_{av_2} denote the average power of $x_1(t)$ and $x_2(t)$ respectively. Called the "correlation" between $x_1(t)$ and $x_2(t)$, the third term in eqn (1) indicates how "similar" these two waveforms are. If generated by independent devices, the noise waveforms are usually "Uncorrelated" and the integral in (1) vanishes. For example, the noise produced by a resistor has no correlation with that generated by a transistor.

On such a case $P_{av} = P_{av_1} + P_{av_2}$. From this observation, we say superposition holds for the power of uncorrelated noise sources.

Types of Noise

(a) Device electronic noise, (b) Environmental noise.

Device Electronic noise:

(a) Thermal Noise, (b) Flicker Noise, (c) Shot noise, (d) Popcorn Noise.

Thermal Noise :-

Resistor Thermal Noise: The random motion of electrons in a conductor introduces fluctuations in the voltage measured across the conductor even if the average current is zero. Thus the spectrum of thermal noise is proportional to the absolute temp.

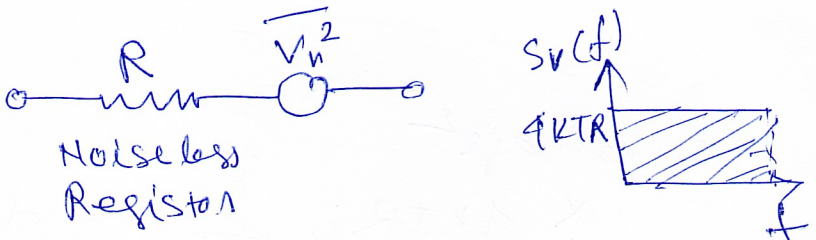


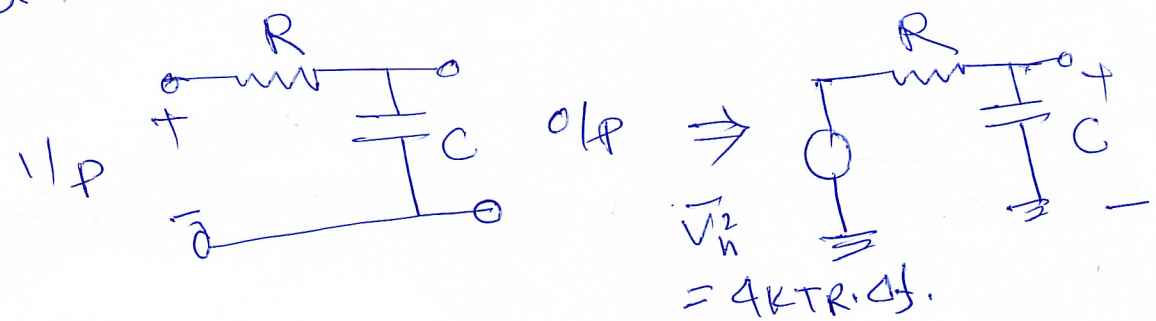
Fig: Thermal noise of resistor

The thermal noise of a resistor R can be modeled by a series voltage source, with the one-sided spectral density.

Now $\overline{V_n^2} = 4KTR \cdot \Delta f$ (Mean square voltage)

where $K = 1.38 \times 10^{-23}$ J/K is the Boltzman constant, and Δf is the bandwidth.

Example: 1 Consider the RC circuit in Fig. Calculate the noise spectrum and the total noise power in V_{out}.



Now voltage across capacitor by ~~tap~~ Laplace transform

$$= V_n(s) \cdot \frac{1}{1+sCR}$$

Putting $s = j\omega$

$$V_{out}(j\omega) = V_n(j\omega) \cdot \frac{1}{1+j\omega CR}$$

or, $$|V_{out}(j\omega)|^2 = |V_n(j\omega)|^2 \cdot \frac{1}{1+\omega^2R^2C^2}$$

$$= \frac{4KTR \cdot d\omega}{2\pi} \cdot \frac{1}{1+\omega^2R^2C^2}$$

Total noise voltage^2

$$\overline{V_{ou}^2} = \int_0^\infty \frac{4KTR}{2\pi} \cdot \frac{1}{1+\omega^2R^2C^2} d\omega$$

$$= \frac{4KTR}{2\pi} \int_0^\infty \frac{d\omega}{1+\omega^2R^2C^2}$$

We know $\int \frac{dx}{1+x^2} = \tan^{-1}x$, Put $x = \omega RC$

$$\frac{dx}{RC} = d\omega$$

$$= \frac{4KTR}{2\pi R} \int_0^\infty \frac{dx}{1+x^2}$$

$$= \frac{4KTR}{2\pi RC} \cdot \left[\tan^{-1} x\right]_0^\infty$$

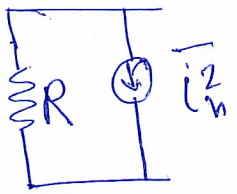
$$= \frac{4KTR}{2\pi RC} \cdot \frac{\pi}{2} = \frac{KT}{C} \quad \cdots (2)$$

Note that the unit of $\frac{KT}{C}$ is V^2. We may also consider $\sqrt{KT/C}$ as the total rms noise voltage measured at the output.

Equation (2) implies that the total noise at the output of the circuit is independent of the value of R. Intuitively, this is because for larger values of R, the associated noise per unit bandwidth decreases while the overall bandwidth of the circuit decreases. The fact that KT/C noise can be decreased by only increasing C (if T is fixed)

* The dependence of thermal noise (and some other types of noise) upon T suggests that low temperature operation can decrease the noise in analog circuits.

The thermal noise of resistor can be represented by a parallel current source as well.

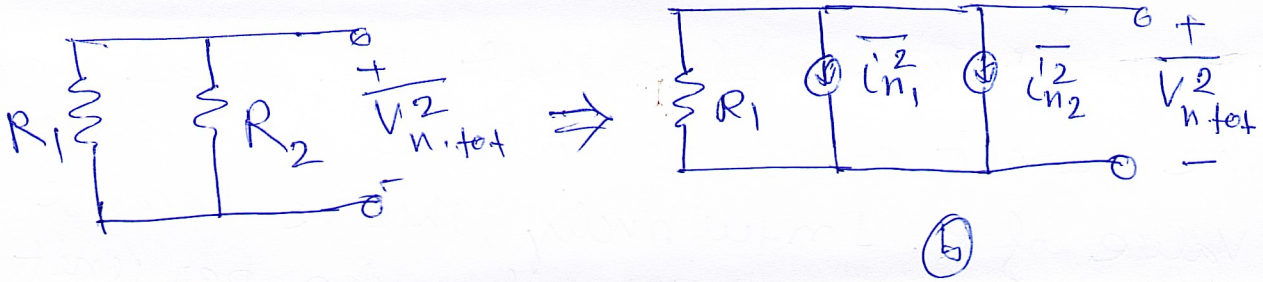


Noiseless Resistor

Note that $\overline{I_n^2}$ is expressed to A^2/Hz.

Fig. Representation of resistor for thermal noise by a current source.

Example: Calculate the equivalent noise voltage of two parallel resistors R_1 and R_2.



(a) (b)

As shown in Fig. b, each resistor exhibits an equivalent noise current with the spectral density $\frac{4KT}{R}\cdot\Delta f$. Since the two noise sources are uncorrelated we add the powers:

$$\overline{i_{n,tot}^2} = \overline{i_{n_1}^2} + \overline{i_{n_2}^2} = 4KT\cdot\left(\frac{1}{R_1}+\frac{1}{R_2}\right)\cdot\Delta f$$

Thus, the equivalent noise voltage is given by

$$\overline{V_{n,tot}^2} = \overline{i_{n,tot}^2}\,(R_1 \| R_2)^2 = 4KT\Delta f\,(R_1 \| R_2)$$

MOSFETs: MOS transistors also exhibit thermal noise. The most significant source ~~in~~ is the noise generated in the channel. It can be proved that for long-channel MOS devices, the channel noise can be modeled by a current source connected between the drain and source terminals with a spectral density: $\overline{i_n^2} = 4KT\gamma g_m$.

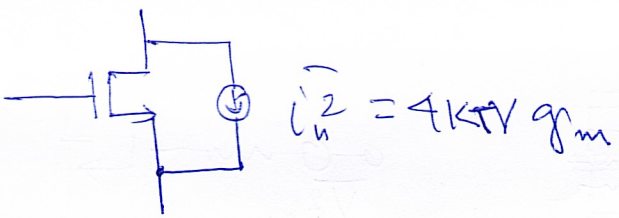


Fig. Thermal noise of a MOSFET

The coefficient γ (not to be confused with the body effect coefficient!) is derived to be equal to 2/3 for long-channel transistors and operating in saturation region. The PSD is

$$\frac{\overline{i_{ds}^2}}{\Delta f} = 4KT(2/3)\cdot g_m = \frac{8}{3}KT g_m$$

and in the triode region $g_{ds} = \frac{\partial I_D}{\partial V_{DS}}$

$$g_{ds} = \frac{\partial}{\partial V_{DS}}\left[\mu_n C_{ox}\frac{W}{L}\left\{(V_{GS}-V_T)\cdot V_{DS} - \frac{V_{DS}^2}{2}\right\}\right]$$

$$= \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T-V_{DS})$$

The thermal noise associate with the

triode region

$$\frac{\overline{i_{ds}^2}}{\Delta f} = 4kT \cdot g_{ds}$$

For submicron MOSFET γ may be replaced by a larger value. For example, γ is 2.5 in some 0.25 μm MOS devices.

The ohmic sections of a MOSFET also contribute thermal noise.

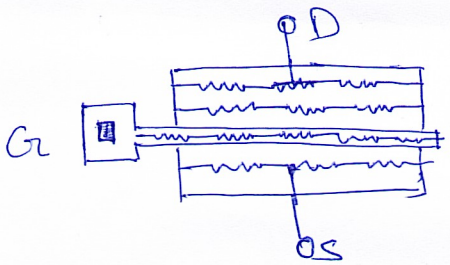


Fig. a Layout of a MOSFET indicating the thermal resistance.

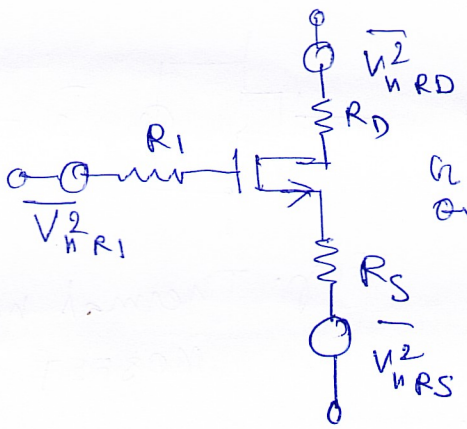
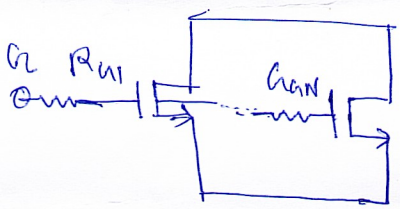


Fig. (b) Circuit model.



$R_{G1} + R_{G2} + \cdots R_{Gn} = R_G$

Fig. c distributed gate resistance

Fig. a, the gate, source, and drain materials exhibit finite resistivity, thereby introducing noise. For a relatively wide transistor, the source and the drain resistance is typically negligible whereas the gate distributed resistance may become noticeable.

In the noise model of Fig. (b), a lumped resistor R_1 represents the distributed gate resistance. Viewing the overall transistor as the distributed structure shown in Fig. c, we observe that the unit transistor near the left end see the noise of only a fraction of R_G whereas those near the right

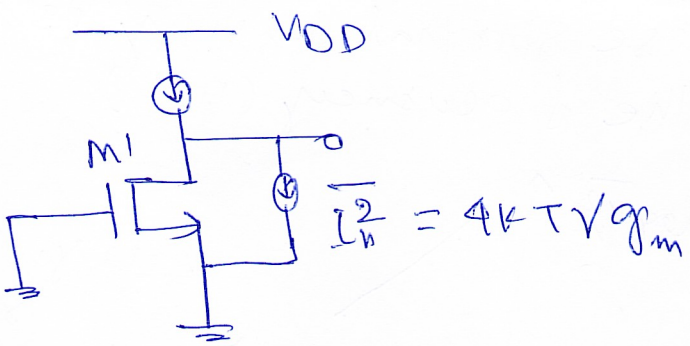
end see the noise of most of R_G. We therefore expect the lumped resistor in the noise model to be less than R_G.

While the thermal noise generated in the channel is controlled by only the transconductance of the device, the effect of R_G can be reduced by proper layout.

Example: Find the maximum noise voltage that a single MOSFET can generate.

Soln: As shown in Fig. the maximum output noise occurs if the transistor sees only its own output impedance as the load. i.e if the external load is an ideal current source. The output noise voltage is then given by $\overline{V_n^2} = \overline{I_n^2} R_o^2$

$= 4kT\left(\frac{2}{3} g_m\right) R_o^2$. ---- equation suggests that



The noise current of a MOS transistor decreases if the transconductance drops. For example if the transistor operates as a constant current source, it's desirable to minimize its transconductance. Another important conclusion is that the noise measured at the output of the circuit does not

depend on where the input thermal is because for output noise calculation, the input is set to zero.

Flicker Noise:- The interface between the gate oxide and the silicon substrate in a MOSFET, many "dangling" bonds appear, giving rise to extra energy states. As charge carriers move at the interface, some are randomly trapped and later released by such energy states, introducing "flicker" noise in the drain current.

The flicker noise is modeled as a voltage source in series with the gate and given by

$$\frac{\overline{V_n^2}}{\Delta f} = \frac{K}{C_{ox} W \cdot L} \cdot \frac{1}{f} \quad \cdots (1)$$

where K is a process-dependent constant on the order of $10^{-25} V^2 F$.

The noise spectral density is inversely proportional to the frequency (Fig.)

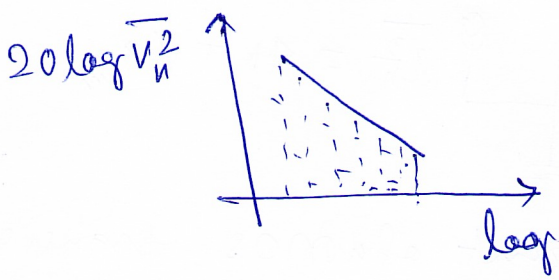


Fig. Flicker noise spectrum.

The trap-and release phenomenon associated with the dangling bonds occurs at low frequencies more often. For this reason flicker noise is also called 1/f noise. Flicker noise does not depend on the bias current or the temperature.

The inverse dependence of on WL suggests that to decrease 1/f noise, the device area must be increased. The PMOS devices exhibit less 1/f noise than NMOS because the former carry the holes in a "buried" channel i.e at some distance from the oxide-silicon interface.

In order to quantify the significance of 1/f noise with respect to thermal noise for a given device, we plot both spectral densities on the same axes (Fig.). Called the 1/f noise "corner frequency," the intersection point serves as a measure of what part of the band is mostly corrupted by flicker noise.

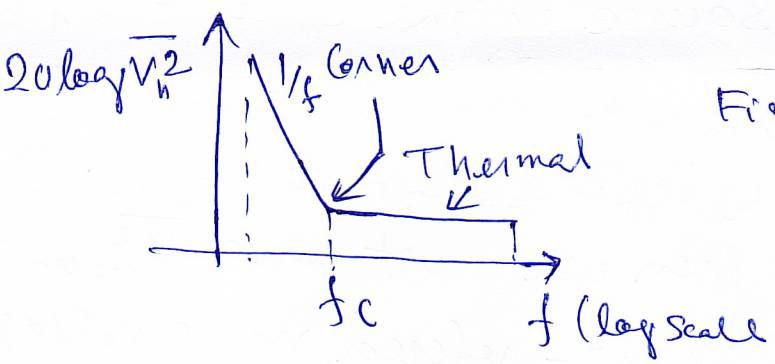


Fig. Concept of flicker noise Corner frequency.

For MOSFET in saturation, the 1/f noise corner, f_c, of the output current is determined as

$$4kT\left(\frac{2}{3}g_m\right) = \frac{K}{C_{ox}WL}\cdot\frac{1}{f_c}\cdot g_m^2$$

that is

$$f_c = \frac{K}{C_{ox}WL}\cdot g_m\cdot\frac{2}{8kT}$$

This result implies that f_c generally depends on device dimensions and bias current.

(i) Flicker noise is "coloured noise", i.e PSD vary with frequency

(ii) Flicker noise is normally present when the transistor has bias current (i,e in the saturation region).

(iii) Because of $1/f$ dependency, the flicker noise is dominant at very low frequency.

Determination of input-refered noise voltage:
In fig(a) the idea is to represent the effect of all noise source in the circuit by a single source, $\overline{V_{n,in}^2}$, at the input such that the output noise (Fig.b) equals that in Fig.(a). If the voltage gain is Av, then we must have $\overline{V_{n,out}^2} = A_v^2 \overline{V_{n,in}^2}$, that is the input-refered noise voltage in this is given by the output noise voltage divided by the gain.

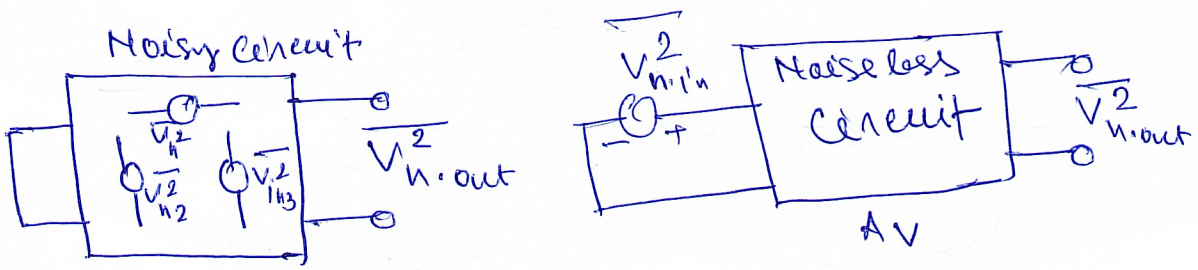
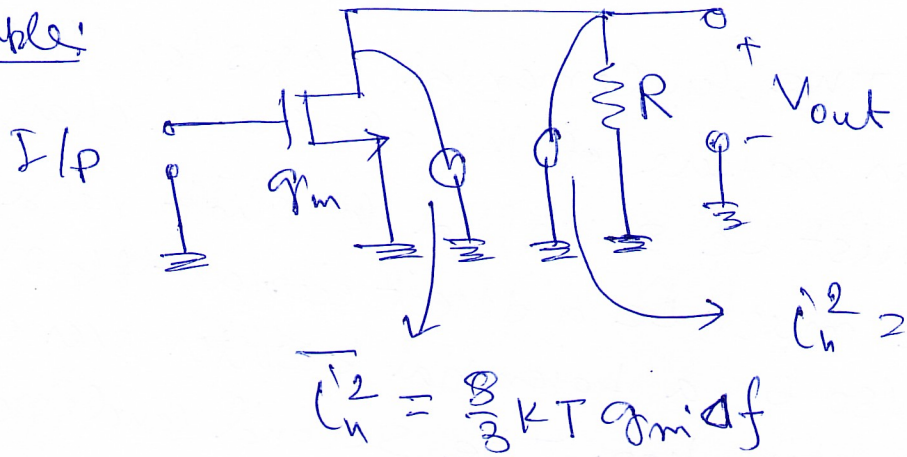


Fig. a

Example:



Total noise is

$$\cancel{V_{out}} \cdot \overline{V_{n.out}^2} = (R \| r_{ds})^2 \left(\frac{4KT}{R} + \frac{8}{3} KT g_m \right) \cdot \Delta f$$

$$= (R \| r_{ds})^2 \left(\frac{1}{R} + \frac{2}{3} g_m \right) 4KT \cdot \Delta f$$

Now what is input refered noise.

The gain of this amplifier $A_V = g_m (R \| r_{ds})$

$$\therefore \overline{V_{n,in}^2} = \frac{\overline{V_{n.out}^2}}{A^2} = \frac{\overline{V_{n.out}^2}}{g_m^2 (R \| r_{ds})^2} \cdot (R \| r_{ds})^2 \left(\frac{1}{R} + \frac{2}{3} g_m \right) 4KT \Delta f$$

$$= \frac{1}{g_m^2} \cdot \left(\frac{1}{R} + \frac{2}{3} g_m \right) 4KT \cdot \Delta f$$

$$= \left(\frac{1}{g_m^2 R} + \frac{2}{3 g_m} \right) 4KT \cdot \Delta f$$

$$= \frac{8}{3} \frac{4KT}{g_m} \cdot \Delta f + \frac{4KT}{g_m^2 R} \cdot \Delta f$$

For Channel — For resistor

Shot Noise: The fundamental basis for shot noise is the granular nature of the electronic charge. Two conditions must be satisfied for shot noise to occur. There must be a direct current flow and there must also be a potential barrier over which the charge carriers hop. The second condition tells us that, linear resistors do not generate shot noise, despite the quantized nature of electronic charge. Now

$$\bar{i}_n^2 = 2qI_{DC}\Delta f.$$

where $\bar{i}_n$ is the rms noise current, q is electronic charge (about 1.6×10^{-19} C), I_{DC} is the DC current in amperes, and Δf is the noise bandwidth in hertz. Note that, like thermal noise, shot noise (ideally) is white and has an amplitude that possesses a Gaussian distribution. As a reference point, the rms current noise density is approximately 18 pA/$\sqrt{\text{Hz}}$ for a 1 mA value of I_{DC}.

* Representation of noise by voltage and current sources:
We model the input-referred noise by both a series voltage source and a parallel current source so that if the output impedance of the preceding stage assumes large values - thereby reducing the effect of $\overline{V_{n,in}^2}$ - the noise current source still flows through a finite impedance, producing noise at the input.

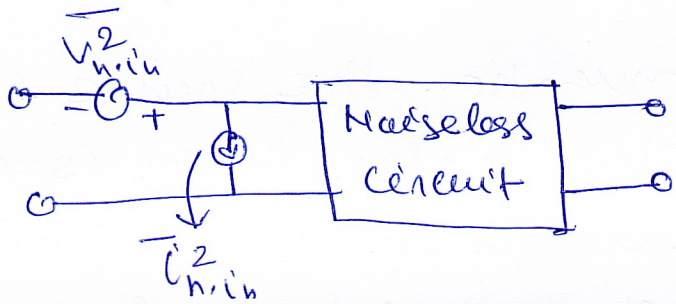


Fig. a Representation of noise by voltage and current sources

How do we calculate $\overline{V_{n,in}^2}$ and $\overline{I_{n,in}^2}$? Since the model is valid for any source impedance, we consider two extreme cases: zero and infinite source impedance. As fig. b, if the source impedance is zero, $\overline{I_{n,in}^2}$ flows through $\overline{V_{n,in}^2}$ and has no effect on the output. Thus, the output noise measured in this case arises solely from $\overline{V_{n,in}^2}$, Similarly, if the input is open (Fig. c), then $\overline{V_{n,in}^2}$ has no effect and the o/p noise is due to only $\overline{I_{n,in}^2}$.

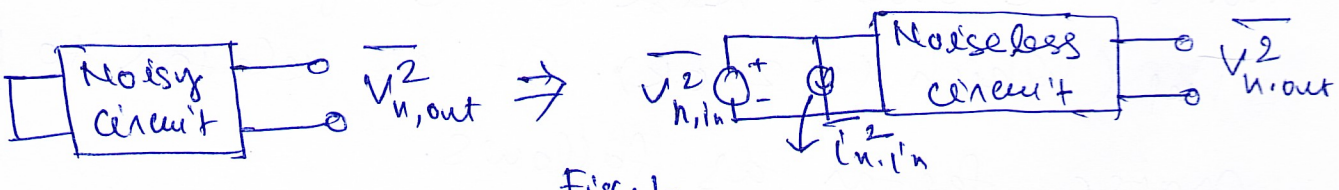


Fig. b

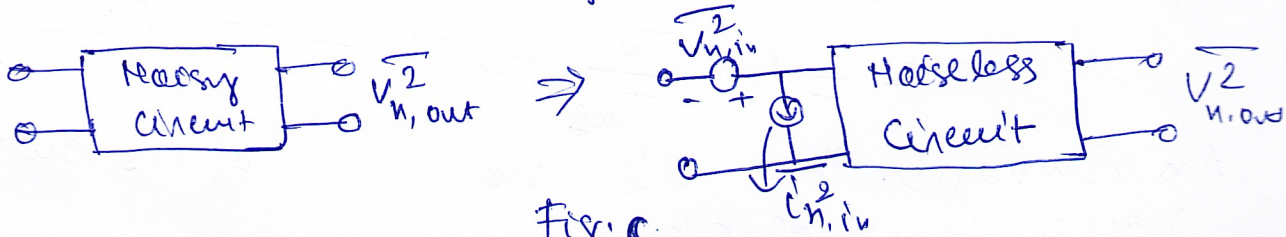


Fig. c

Noise Factor: A useful measure of the noise performance of a system is the noise factor, denoted F.

The noise factor is defined as

$$F \equiv \frac{\text{total output noise power}}{\text{output noise due to input source}}$$

where, by convention, the source is at a temp of 290K. The noise factor is a measure of the degradation in signal-to-noise ratio that a system introduces.

Noise Figure and Noise temperature:- The noise figure (NF) is simply the noise factor expressed in decibels.

Noise temperature, T_N is an alternative way of expressing the effect of an amplier's noise contribution and is defined as the increase in temperature required of the source resistance for it to account for all of the output noise at the reference temperature T_{ref} (which is 290K). It is related to the noise factor as follows.

$$F = 1 + \frac{T_N}{T_{ref}} \Rightarrow T_N = T_{ref}(F-1).$$

An amplifier that adds no noise of its own has a noise temp of 0K.